

Disease Analysis Dashboard Report

Sl No.	Contents	Page No.
1.	Introduction	4
2.	Aim of this Project	5
3.	Description of The Dataset	6
4.	Preparing The Dataset	7
5.	Main Dashboard	8
6.	Visualisation 1 - Classification by Disease Type	9
7.	Visualisation 2 - Heart Attack Rate by Smoking Status and Gender	10
8.	Visualisation 3 - Heart Attack Map Chart	11
9.	Visualisation 4 - Average BMI by Disease, Gender and Physical Activity	12
10.	Visualisation 5 - What If :- Adjusted Heart Attack Risk by Age & Gender	13
11.	Interactive Features in The Dashboard	14
12.	Calculated Fields in The Dashboard	15
13.	What If Analysis	16
14.	Ethical Usage	17
15.	Conclusion	18
16.	Recommendations	19
17.	Appendices	20

- **Glossary Of Terms:**

- Heart Attack- A condition that damages the heart muscle by obstructing blood flow to the heart.
- Heart Attack Rate- The proportion of people in the dataset who have had a heart attack.
- Adjusted Heart Attack Risk- A risk number that has been altered to account for various scenarios based on the risk multiplier.
- A user-controlled parameter used in what-if analysis to raise or lower the risk of a heart attack is called a risk multiplier.
- Body Mass Index - or BMI, is a metric used to evaluate body fat levels based on height and weight.
- BMI Rounded - For easier display, BMI numbers are simplified.
- Smoker Status - Indicates a person's status as a regular, occasional, former, or non-smoker.
- Physical Activity - Indicates whether a person works out on a regular basis.
- Age Category - For simpler analysis, age ranges are grouped (e.g., 18–24, 25–29).
- Disease Type - Groupings of conditions including diabetes, asthma, COPD, stroke, and so on.
- A heatmap - is a chart that illustrates values and patterns using colour intensity.
- Dashboard - An interactive visual representation that combines several filters and charts.
- Data cleaning - is the process of preparing data by standardising formats and addressing missing values.
- Calculated Field - A new variable built with formulas from preexisting data.
- Data visualisation - is the process of displaying data graphically in order to spot patterns and insights.

- **Data Dictionary**

Field Name	Description	Example Values
Age Category	Age group of individuals	18-24, upto 80+
Sex	Gender	Male, Female
Smoking Status	Smoker / Non- Smoker	Yes, No
BMI	Body Mass Index	25.00 - 34.00
Physical Activity	Level of Physical Activity	Low, Moderate, High
Disease Type	Primary disease category	Arthritis, Asthma, Diabetes, COPD, Skin Cancer, Angina, Kidney, Stroke, Depressive Disorder
Heart Attack	Binary indicator of whether the individual has had a heart attack	0, 1 (No, Yes)
Heart Attack Rate	Individuals in a group who had a heart attack	0.05, 0.12, 0.18, etc
Adjusted Heart Attack Rate	Heart attack rate adjusted using a risk-multiplier parameter (What-If)	0.5 - 1.5
State	US state associated with the number of Individuals with Heart Attack.	New York, California, Texas, etc

- Introduction :

This project analyses a large population health dataset using sophisticated interactive data visualisation tools. The dataset includes a wide range of characteristics, including clinical indicators associated with different health disorders, lifestyle indicators such as body mass index (BMI), physical activity levels, and smoking status, and demographic information including age and gender. The project's goal is to better understand how behavioural, demographic, and medical factors affect overall health outcomes across populations by identifying significant correlations between them through the integration of these several data dimensions.

The analysis's focus is expanded to cover a number of chronic and lifestyle-related illnesses, including diabetes, asthma, kidney disease, stroke, and arthritis, rather than only heart attack risk. Since many of these diseases share risk factors and frequently co-occur, examining these ailments collectively enables a more comprehensive understanding of population health. By identifying overlapping patterns and interdependencies, this multi-disease approach provides a deeper understanding of how combinations of risk variables contribute to illness prevalence and development among various groups.

An interactive Tableau dashboard was created to illustrate important patterns and trends in the dataset in order to facilitate this research. The dashboard's many visual components are intended to draw attention to the distribution of diseases, lifestyle-related risk factors, regional variances, and population-specific BMI disparities. Users may easily spot differences and correlations that might not be immediately apparent in raw data thanks to the way these visualisations are designed to encourage natural exploration.

The inclusion of a "what-if" analysis, which enables users to model changes in particular risk factors and see the possible influence on health outcomes, is a crucial component of the project. By shifting from static reporting to scenario-based exploration, this predictive component improves the dashboard's analytical capabilities. Users may instantly observe how changes in variables like physical activity levels or smoking rates affect patterns of illness risk.

A key component of the dashboard's value is its interactivity. Users can customise the analysis to target specific groups of interest by applying filters based on state, age group, gender, or particular health issues. The visualisations update instantly when filters are changed, facilitating exploratory investigation and provide prompt feedback. The dashboard's dynamic flexibility turns it from a straightforward reporting tool into an interesting platform for study and instruction. In addition to describing current trends, it enables users to decipher underlying patterns and forecast population-level health effects.

- Aim of this Project:

This project's objective is to examine population health data and find connections between lifestyle, disease-related, and demographic factors. The dashboard is intended to offer information on geographic trends, BMI patterns, heart attack risk, and disease prevalence.

Creating an interactive environment where users can compare groupings, filter data, and conduct what-if analysis is another goal. This aids in comprehending how variations in risk factors may impact health outcomes for various age groups and genders.

Project Objectives.

- Describe the population-health dataset's variables and structure.
- Make derived indicators and handle missing values as you clean and prepare the dataset for Tableau.
- With five key visualisations and interactive controls, create a cohesive, multi-view dashboard.
- Examine each visualisation in depth, analysing the correlations between the variables.
- Using the Risk Multiplier parameter, talk about the "What-If" analysis.
- Describe how the dashboard's interactive filters and actions operate.
- When visualising health-related data, especially if it is simulated, keep ethics in mind.
- Summarise the main conclusions and include recommendations on how the dashboard might be expanded for use in actual research or learning environments.

- Description of The Dataset :

This project's dataset is a simulated population health dataset. Along with lifestyle factors like smoking status, physical activity, BMI, and sleep patterns, it also contains demographic characteristics like age group, gender, and geographic region. Numerous disease indicators, including diabetes, asthma, arthritis, COPD, kidney disease, and stroke, are also included in the collection.

The dataset contains both numerical and category variables, enabling the creation of many kinds of visualisations. This enables the identification of trends in various diseases and lifestyle factors as well as comparisons between population groupings.

The dataset was selected because it facilitates multidimensional analysis and enables investigation of the connections between risk variables and health problems. Additionally, it makes interactive visualisation possible through what-if analysis, mapping, and filtering. The dataset's primary variables are:

1. Age Group, Gender, and Smoking Status - The impact of age, gender, and smoking behaviour on disease prevalence and heart attack risk in various population groups was examined using these demographic and lifestyle factors.
2. BMI and Exercise - In order to investigate the connection between lifestyle choices and general health, especially with regard to illness conditions and risk levels, body mass index and physical activity levels were examined.
3. Hours of Sleep - To investigate its possible influence on general welfare and its correlation with disease patterns within the dataset, sleep duration was incorporated as a health indicator.
4. Mental Health Days Illnesses - For understanding its relationship with physical health issues and general population health trends, the number of illness days related to mental health was taken into consideration.
5. Geographical State Heart Attack Indicator - To find regional trends and differences in heart attack instances among several states, geographic location and the frequency of heart attacks were examined jointly.

- Preparing The Dataset :

The dataset was prepared and cleaned before the visualisations were made. Categorical variables were standardised and missing values were addressed. To facilitate analysis, a number of computed fields were developed.

The following adjustments were made:

1. Elimination of null values - To guarantee data accuracy and prevent mistakes in analysis and visualisation, missing or null values were found in the dataset and either eliminated or treated appropriately.
2. The establishment of age groups - To make analysis easier and enable more transparent comparisons across various age ranges, age values were categorised into relevant groups, such as 18–24, 25–34, and so forth.
3. Standardisation of labelling for diseases - To guarantee uniformity throughout the collection, disease names were standardised. This enhanced picture clarity and prevented redundancy brought on by multiple spellings or formats.
4. Development of a heart attack signal - To make it simpler to calculate heart attack rates and compare between groups, a binary variable was developed to indicate whether a person had a heart attack.
5. Heart attack rate calculation - In order to provide a clear indicator for risk analysis, a computed field was created to determine the percentage of people in each group that had heart attacks.
6. Establishing a modified risk metric - A risk multiplier parameter was used to generate an adjusted heart attack risk metric that allowed what-if analysis to simulate changes in health risk under various situations.
7. BMI rounded value creation - In order to simplify the data and enhance readability in visualisations like heatmaps, BMI values were rounded, making patterns simpler to understand.
8. Establishment of illness category classification - To make it easier to compare and analyse health situations across the population, diseases were categorised into more expansive groups.

To facilitate sophisticated visualisations, a number of computed fields were developed. To show the percentage of people who had heart attacks, a heart attack rate measure was computed. To enable what-if analysis, an adjusted heart attack risk field was also created using a risk multiplier parameter.

To make the heatmap presentation easier to read, BMI values were rounded. These changes made it possible to use the dataset efficiently for dashboard development and interactive visualisations.

- Main Dashboard :

The dashboard analyses population health and illness risk from several angles by combining a number of visualisations. These include the distribution of diseases, the effects of smoking, regional trends, BMI comparisons, and risk simulations.

Every chart is interactive and linked via parameters and filters. Users can interactively explore the data by filtering by gender, disease kind, and other characteristics. It is possible to simulate higher or lower risk levels using the risk multiplier parameter.

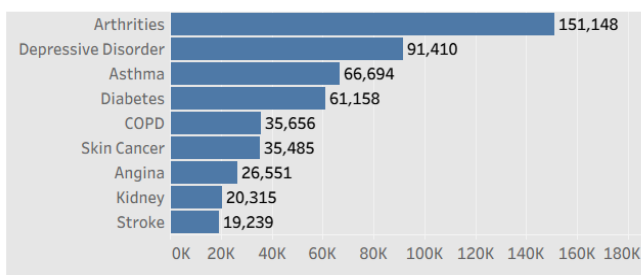
This dashboard facilitates exploratory data analysis and offers a thorough overview of health trends. Users can compare disease prevalence, identify high-risk groups, and look at lifestyle factors that impact health outcomes.

The dashboard consists of:

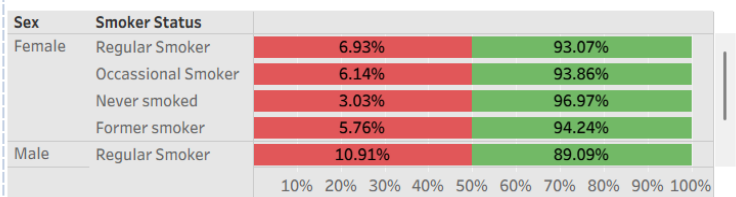
1. Number of People by Disease Type Heart Attack Risk by Gender and Smoking Status.
2. Geographic Distribution Map.
3. Average BMI by Physical Activity and Illness.
4. What-If Gender and Age-Adjusted Heart Attack Risk.

Disease Analysis Dashboard

Classification by Disease Type



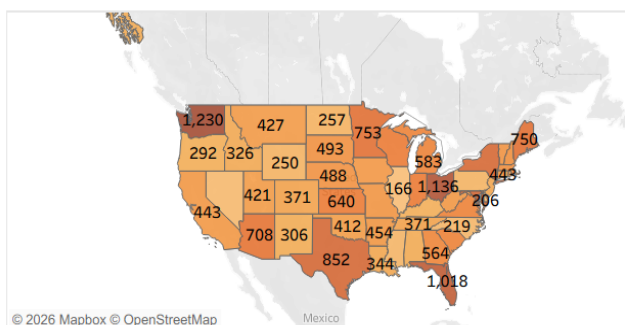
Heart Attack Rate by Smoking Status and Gender



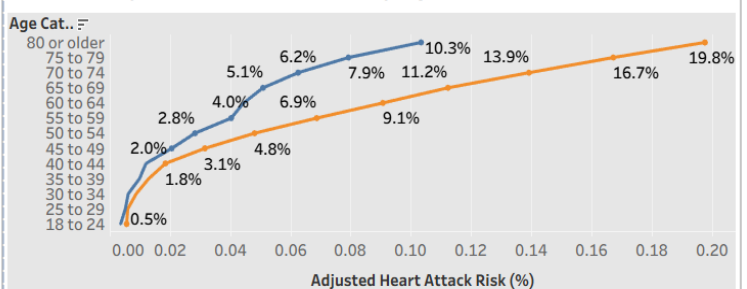
Average BMI by Disease, Gender and Physical Activity

Disease_Type	Sex / Physical Activities			
	Female		Male	
	No	Yes	No	Yes
Diabetes	34.000	31.000	32.000	31.000

Heart Attack Map Chart



What If :- Adjusted Heart Attack Risk by Age & Gender

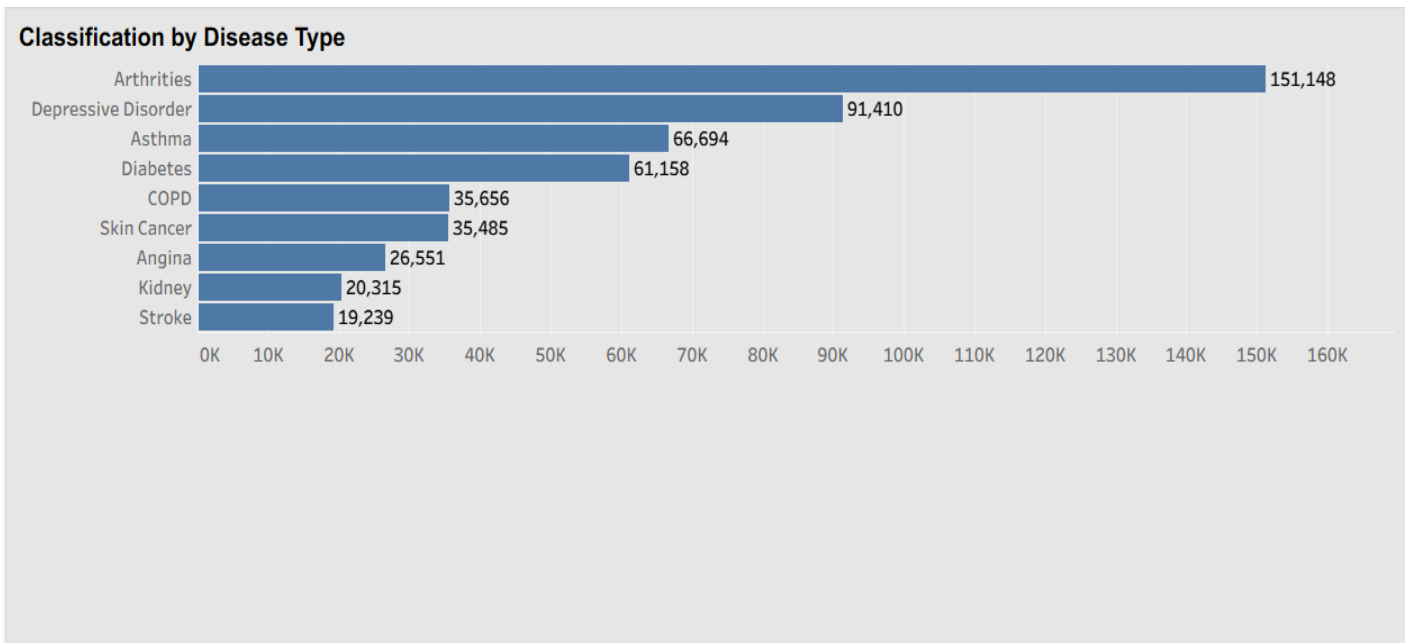


- Visualisation 1 - Classification by Disease Type :

The number of people impacted by each illness is displayed in this bar graph. It gives a general picture of the population's disease prevalence. Asthma, depression, and arthritis are more common conditions, whereas renal disease and stroke are less common.

The most prevalent medical conditions in the dataset can be found using this visualisation. Additionally, it makes it possible to compare various diseases and determine how they affect the population in relation to one another.

According to the graphic, chronic diseases are common and may necessitate targeted healthcare planning.

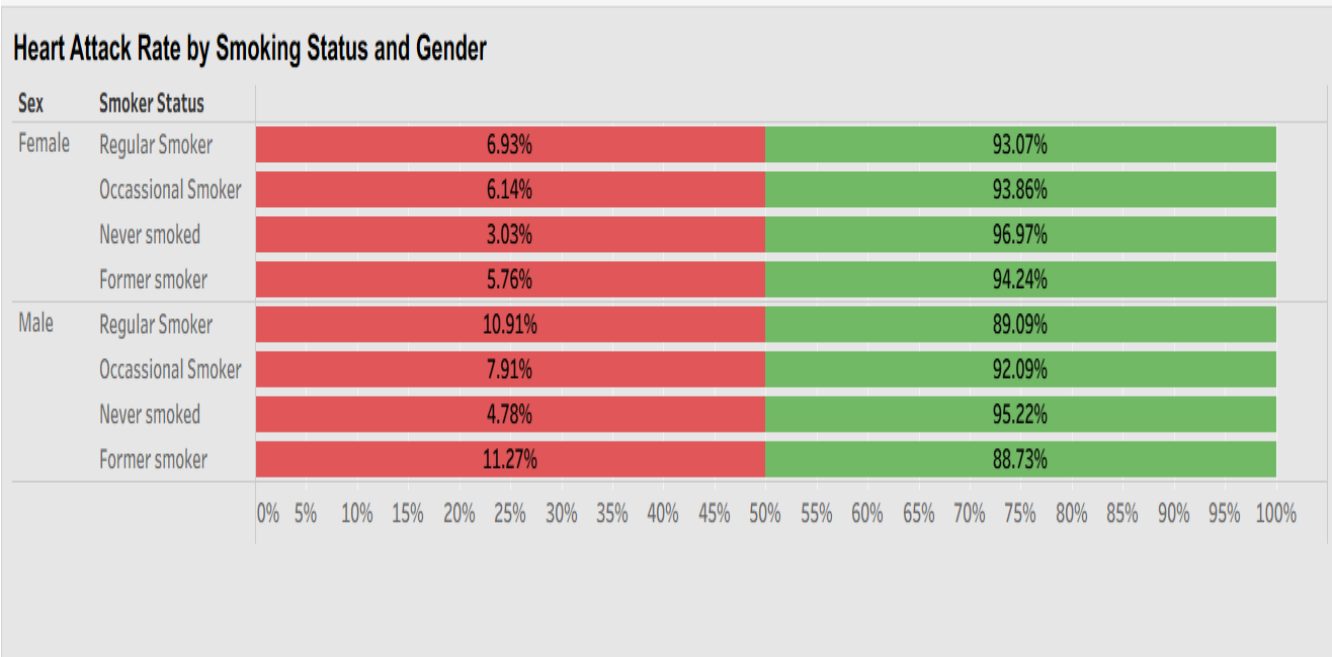


- Visualisation 2 - Heart Attack Rate by Smoking Status and Gender:

Heart attack rates by gender and smoking status are shown in this 100% stacked bar graph. The graphic illustrates the disparities between the male and female populations and contrasts smokers with non-smokers.

Heart attacks are more common in regular smokers than in non-smokers. Furthermore, in the majority of smoking categories, men typically display higher risk levels. This suggests that the risk of a heart attack is influenced by both gender and smoking.

This graphic illustrates how lifestyle choices and health consequences are related.

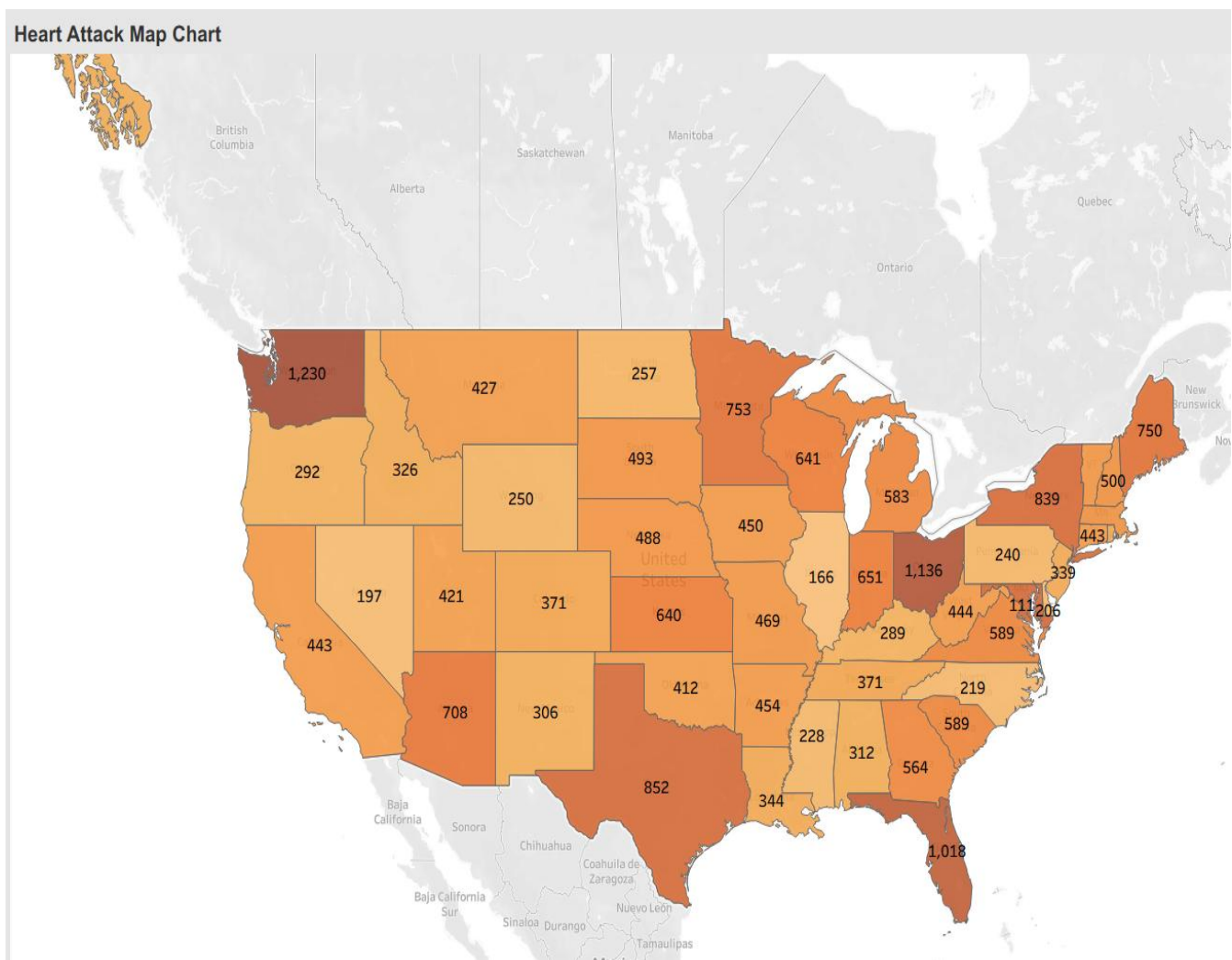


- Visualisation 3 - Heart Attack Map Chart:

The geographical distribution of heart attack cases is shown in this map representation. High-risk zones can be identified by shading each region according to the number of cases.

Heart attack cases are more prevalent in some areas than in others. Population size, lifestyle choices, and other regional characteristics may have an impact on this geographic heterogeneity.

The map facilitates location-based health analysis and aids users in identifying regional patterns.



- Visualisation 4 - Average BMI by Disease, Gender and Physical Activity:

The average BMI values for each disease type, gender, and degree of physical activity are displayed in this heatmap. The graph makes it possible to compare BMI under various circumstances.

People with asthma and diabetes have higher BMI values. People with skin cancer and stroke have lower BMI values. Physically active people typically have lower BMI readings.

This graphic illustrates the connection between disease, lifestyle choices, and BMI.

This graphic provides an answer to the query:

What differences exist in BMI between diseases and levels of physical activity?

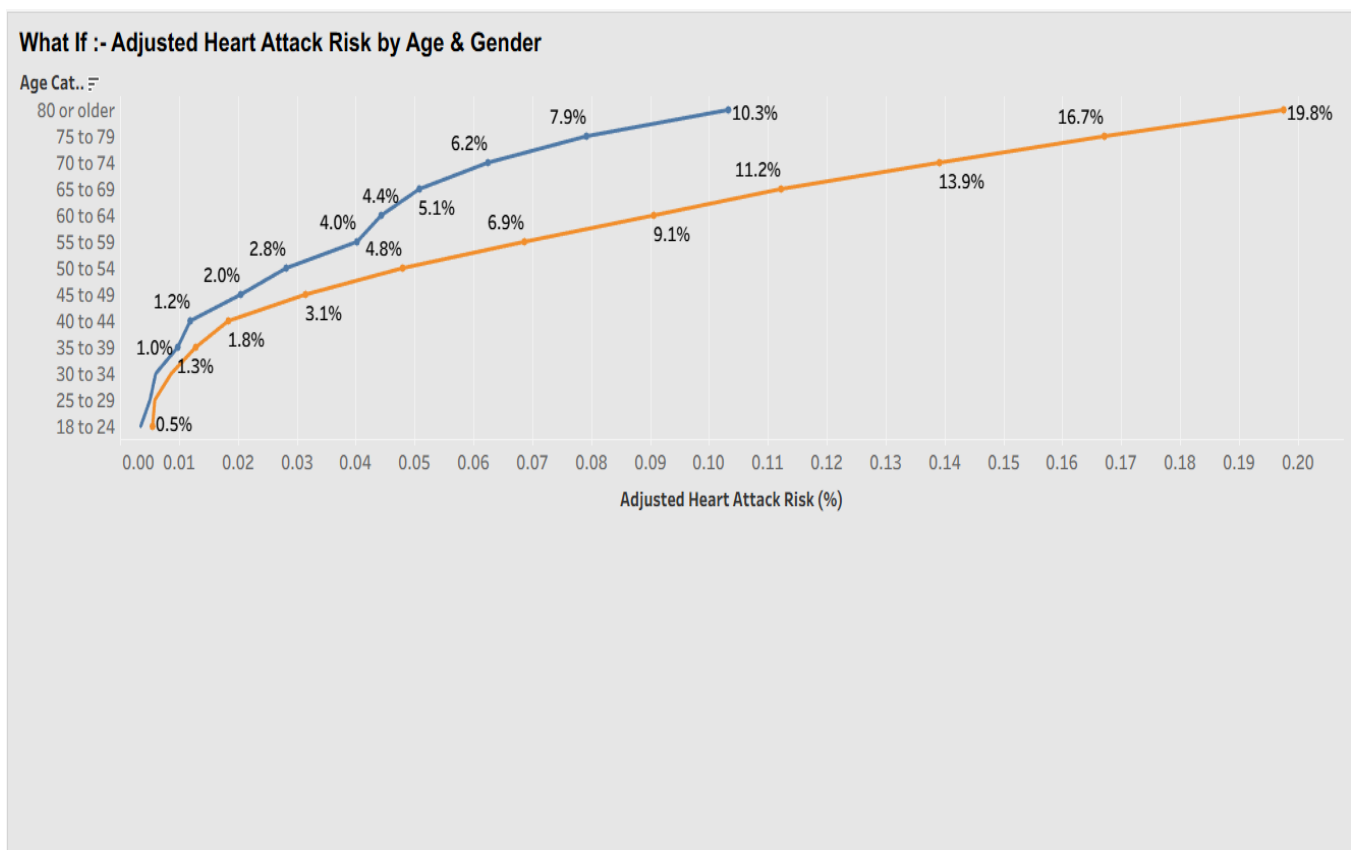
Average BMI by Disease, Gender and Physical Activity				
Disease_Type	Sex / Physical Activities			
	Female		Male	
	No	Yes	No	Yes
Diabetes	34.000	31.000	32.000	31.000
Asthma	32.000	29.000	30.000	28.000
Arthritis	30.000	28.000	30.000	29.000
Kidney Disease	30.000	28.000	30.000	28.000
Depressive Disorder	31.000	28.000	29.000	28.000
COPD	30.000	28.000	29.000	28.000
Stroke	29.000	27.000	29.000	28.000
Angina	28.000	27.000	29.000	28.000
Skin Cancer	28.000	25.000	28.000	27.000

- Visualisation 5 - What If :- Adjusted Heart Attack Risk by Age & Gender:

This line graph uses a risk multiplier parameter to show adjusted heart attack risk by gender and age group. Users can simulate increases or decreases in risk using the slider.

The graph indicates that both genders' chance of having a heart attack rises with age. In general, men are slightly more at risk than women. The adjusted risk increases for all age groups as the multiplier grows.

Users can investigate how changes in risk factors might affect population health outcomes using this what-if analysis.



- Interactive Features in The Dashboard:

To assist users in dynamically exploring the simulated dataset, the dashboard makes use of a number of interactive elements. Changing any of the filters for Gender, Smoking Status, and Disease Type instantaneously updates every chart on the dashboard. Selecting "Male" or "Diabetes," for instance, reshapes the bar chart, heatmap, stacked bar, and map around that subgroup, making it simple to examine trends across various ailments and demographics.

What-if analysis is supported with a Risk Multiplier parameter (slider). The multiplier can be adjusted by users (for example, from 0.5 to 2.0) to model how heart-attack risk varies under various conditions, such as better lifestyles or worsening risk factors. This multiplier helps users see the effects of possible interventions by altering the Adjusted Heart Attack Rate in the line chart as well as the map and stacked bar chart.

Additionally, the dashboard employs linked highlighting: tooltips offer on-hover insights, and clicking on an illness bar, state, or smoking group highlights the same section in other charts. When combined, these elements improve the dashboard's usability, encourage more in-depth investigation of the data's relationships, and assist students in comprehending the application of interactive dashboards in public health studies.

- Calculated Fields in The Dashboard:

Several calculated fields are used in the project:

1. Rate of Heart Attacks

This field determines the percentage of people in each group (e.g., by disease, sex, smoking status, or state) who have experienced a heart attack. It is calculated as

Heart Attack Rate = Number of individuals with Heart Attack /Total number of individuals in group

Instead of relying just on raw counts, this rate serves as the base measure in the 100% stacked bar chart, the map, and the line chart, enabling users to compare heart-attack risk across various subgroups.

2. Modified Risk of Heart Attacks

This field uses a Risk Multiplier parameter to increase the base Heart Attack Rate:

$[\text{Heart Attack Rate}] * [\text{Risk Multiplier}] = \text{Adjusted Heart Attack Rate}$

Users can simulate various scenarios with the Risk Multiplier, a user-controlled slider (e.g., ranging from 0.5 to 1.5):

Reduced risk is indicated by a multiplier less than 1.0 (for example, as a result of healthier behaviours or interventions).

Increased risk is indicated by a multiplier greater than 1.0, such as when obesity or smoking rates grow.

In addition to being utilised in the What-If line chart, the Adjusted Heart Attack Rate can be fed into the map and stacked bar chart when the user want to investigate how risk might alter under fictitious circumstances.

3. BMI Rounded Risk Multiplier Parameter for Disease Type

The continuous BMI data are converted into an additional BMI-based categorical variable to facilitate interpretation in the heatmap and other representations. This field classifies people into groupings like: "Normal", "Overweight", "Obesity"

Cross-category labels are then created by combining these rounded BMI groups with Physical Activity and Disease Type. This enables the dashboard to display:

How each disease's risk patterns vary depending on a person's BMI.

When combined, these computed variables enable users to evaluate risk levels, identify structure in the data, and do what-if analysis without having to manually recalculate statistics outside of Tableau.

Advanced analysis and visualisation are supported by these computations.

- What If Analysis:

Users can model changes in the risk of a heart attack using the what-if analysis. The estimated risk of a heart attack is modified by the risk multiplier parameter.

The risk of a heart attack rises as the multiplier does. Risk decreases when the multiplier drops. This enables users to investigate possible future situations.

Beyond descriptive analysis, this feature offers more insight.

The lines on the What-If chart climb and the map and stacked bar chart show higher simulated heart-attack rates as the user moves the Risk Multiplier slider upward, increasing the adjusted risk proportionately. This may indicate a situation in which risk factors—such as increased rates of smoking, rising obesity, or decreased physical activity—worsen. On the other hand, the adjusted risk drops as the user reduces the multiplier below 1.0, mimicking the effects of effective public health initiatives or improved lifestyle patterns.

This function enables users to investigate potential future scenarios in a methodical manner:

- By lowering the multiplier, they can estimate the answer to the question, "What would happen if smoking prevalence dropped by 30%?"
- Alternatively, they might adjust the multiplier for specific age-sex groups to test the question, "How much would risk increase if obesity levels rose in older adults?"

The What-If feature gives the dashboard a predictive and exploratory layer in addition to basic descriptive analysis (e.g., displaying existing risk patterns). It challenges users to consider cause-and-effect links, contrast best-case and worst-case situations, and get a better understanding of how little adjustments to population-level risk factors can result in significant changes in the burden of heart attacks. This turns the dashboard into a learning tool for thinking like a public health analyst or data-driven policymaker in an educational setting, rather than merely a reporting tool.

- Ethical Usage:

Even when the dataset is simulated, using data visualisation with health-related information creates a number of significant ethical issues. This project's dashboard shows the connections between lifestyle parameters (BMI and physical activity), disease outcomes, and demographic factors (age, sex, and smoking status). These subjects are delicate, therefore care must be taken to prevent misunderstandings or unintentional harm.

First, it's crucial to read things fairly and accurately. Even if the dataset utilised here is fictitious and not based on actual patients, the manner the findings are presented might nevertheless affect viewers' perceptions of actual health problems. False perceptions regarding risk can be produced by exaggerating associations (e.g., suggesting a much stronger effect of smoking than the evidence actually supports) or by employing deceptive scales, colours, or chart kinds. Therefore, the goal of visualisations should be to accurately depict patterns through the use of consistent scales, unambiguous baselines, and suitable chart types that correspond to the underlying data structure.

Second, consumers need to understand that correlation does not imply causality. For example, the dashboard might indicate that smokers are more likely to have heart attacks or that a higher BMI is linked to specific diseases. This does not, however, imply that BMI or smoking are the only factors contributing to those outcomes; a variety of other factors, such as genetics, socioeconomic status, access to healthcare, and environmental impacts, may be involved.

Third, it is important to consider the possibility of stigmatising particular demographic groupings. For instance, a graphic that indicates that a specific age group, sex, or state has a higher illness burden or risk shouldn't be presented in a way that shames or places blame on those groups. Rather, the visualisation ought to stimulate contemplation regarding structural and systemic elements (such access to healthcare or environmental circumstances) that could be responsible for those trends. It is important to avoid perpetuating preconceptions by using neutral, nonjudgmental language when writing labels, titles, and tooltips.

Lastly, ethical data visualisation is supported by computation and labelling transparency. The dashboard describes the function of the Risk Multiplier in the "What-If" analysis and provides comprehensive documentation of how important metrics, including the Heart Attack Rate and Adjusted Heart Attack Rate, are computed. In order to make it clear that the dataset is simulated and that the results are intended for educational or illustrative reasons rather than as exact estimates of real-world risk, axis labels, chart titles, and tooltips are created. This transparency encourages a more responsible and critical attitude to health data and aids users in appropriately interpreting the graphics.

- Conclusion:

This project used interactive visualisations to analyse population health data. Relationships between disease prevalence, lifestyle factors, BMI, and heart attack risk were shown on the dashboard. Higher BMI, smoking, and advanced age were linked to higher risk.

While the heatmap displayed variations in BMI across diseases and exercise levels, the geographic visualisation displayed geographical variance in heart attack cases. The what-if analysis gave more insight into possible outcomes and made it possible to simulate changes in risk.

All things considered, the dashboard offers an interactive tool for investigating health patterns and bolstering data-driven insights.

By showing changes in average BMI across various diseases and levels of physical activity, the heatmap performs a crucial function. For instance, data shows that those with higher BMIs, particularly men and those with poor levels of physical activity, are more likely to have illnesses including diabetes, kidney disease, and depression. This graphic depiction makes it simple to quickly compare disease profiles and determine which subgroups are most burdened by BMI.

In the meantime, the geographic visualisation (map) highlights states or regions with comparatively higher or lower risk by displaying regional heterogeneity in heart attack cases. This helps show how danger may concentrate in specific areas, whether due to simulated variations in smoking rates, age distribution, or lifestyle choices.

The Risk Multiplier parameter drives the What-If analysis, which gives the dashboard a dynamic, forward-looking aspect. The heart-attack rate line chart and associated views are updated in real time by the multiplier, which allows users to mimic higher or lower risk situations. This enables them to go beyond simple description to exploratory forecasting and investigate how changes in smoking, physical activity, or other risk variables can impact future results.

All things considered, the dashboard provides a user-friendly, interactive tool for examining health trends in the population under simulation. It promotes data-driven thinking and an understanding of how geographic, behavioural, and demographic factors interact to influence health risk. The dashboard shows how interactive visualisation can transform complicated health data into understandable insights, even though the data is fictitious.

- Recommendations:

1. Encourage exercise to lower BMI levels.

The dashboard emphasises how a higher BMI is closely linked to a number of chronic illnesses, such as diabetes, renal disease, and depression, particularly when combined with inactivity. Therefore, to help populations lower their BMI and lower their risk of metabolic and cardiovascular illnesses, public health initiatives should encourage regular physical activity (such as walking programs, workplace fitness programs, and community-based sports).

2. Promote initiatives aimed at reducing smoking.

The graphics highlight the harmful effects of tobacco use on cardiovascular health by demonstrating that smokers routinely have greater heart attack rates than non-smokers. These results can be used by policymakers and health organisations to improve smoking-cessation initiatives like public awareness campaigns and subsidised nicotine replacement treatments.

3. Pay attention to age groups at high risk.

Heart attack risk increases significantly in middle and older age groups, especially between 50–59 and 60–69, according to the line graph and age-group studies. This implies that preventive healthcare initiatives, such as routine cardiovascular screening, blood pressure and cholesterol monitoring, and early lifestyle modifications, should give priority to these age groups.

4. Keep an eye on areas where diseases are more common.

The global map shows that various chronic diseases and heart attack risk are concentrated in particular simulated locations, which may be indicative of underlying environmental, societal, or economic risk factors. Such visualisations can be used by public health authorities to pinpoint high-risk areas and provide resources like mobile clinics, specialised outreach, and focused preventative campaigns to those areas.

5. Make use of predictive analytics to plan for future health.

The What-If analysis shows how predictive thinking can help with future planning: by using the Risk Multiplier to simulate higher or lower risk levels, users may understand how various scenarios might impact the burden of disease and heart attacks years in the future. This promotes the application of predictive analytics in health planning, such as predicting the effects of policies aimed at reducing smoking, efforts to avoid obesity, or trends in the ageing population. Tools like this dashboard, when paired with real-world data, can assist governments, hospitals, and insurers in more proactive and evidence-based resource allocation, preventative strategy formulation, and future healthcare demand preparation.

- References:

Pytlak, K. (2022). Indicators of Heart Disease (2022 UPDATE). [online] Kaggle.com. Available at: <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease?select=2022>.

- Appendices:

Appendix A: Raw dataset

1	State	Sex	GeneralHealth	PhysicalHealth	MentalHealth	LastCheckPhysical	AcSleepHours	RemovedToHospital	HadHeartDisease	HadAngina	HadStroke	HadAsthma	HadSkinCancer	HadCOPD	HadDepression	HadKidneyDisease	HadArthritis	HadDiabetes	DeafOrHearingImpaired	BlindOrVisuallyImpaired	DifficultyCaringForSelf	DifficultyWalking	DifficultyDoingHousework	DifficultyEating	SmokerStatus	ECigaretteUse	ChestScar	RaceEthnicity	AgeCategory	Height
2	Alabama	Female	Very good	0	0	Within pas No	8	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	Never smc	Not at all	No	White only	Age 80 or older	
3	Alabama	Female	Excellent	0	0	No	6	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	Never smc	Never use	No	White only	Age 80 or c	
4	Alabama	Female	Very good	2	3	Within pas Yes	5	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Never smc	Never use	No	White only	Age 55 to 5	
5	Alabama	Female	Excellent	0	0	Within pas Yes	7	No	No	No	Yes	No	No	No	No	No	Yes	No	No	No	No	No	No	No	Current sn	Never use	Yes	White only, Non-Hispi		

Appendix B: Cleaned dataset

heart_disease_raw_dataset.csv	heart_disease_raw_dataset.csv	heart_disease_raw_dataset.csv	heart_disease_raw_dataset.csv	heart_disease_raw_dataset.csv	heart_disease_raw_dataset.csv
State	Sex	General Health	Physical Health Days	Mental Health Days	Last Ch
Oregon	Female	Good	15.0000	0.0000	Within p
Utah	Female	Good	6.0000	6.0000	Within p
Virginia	Female	Fair	15.0000	0.0000	Within p
Ohio	Female	Good	2.0000	0.0000	Within p
Texas	Female	Poor	20.0000	15.0000	Within p
Oregon	Female	Fair	30.0000	15.0000	Within p
New York	Female	Poor	7.0000	0.0000	Within p

Appendix C: Selection Process for Visualisation

Characteristics of the data were taken into consideration when choosing the visualisations. To compare disease frequencies, a bar chart was used because it makes it easy to compare different groups. The percentage of heart attack risk across smoking groups and gender were analysed using a 100% stacked bar chart. Since a map offers an intuitive knowledge of regional distribution, it was chosen to visualise geographic patterns. BMI was compared across several variables at once using a heatmap, which made it simple to spot trends. Lastly, trends across age groups and the effect of the risk multiplier parameter were illustrated using a line chart for what-if analysis.

Appendix D: Data Dictionary - Page 3

Appendix E: Inspiration and Criticism for Visualisation

Standard data visualisation concepts including uniformity, simplicity, and clarity have an impact on the dashboard's design. Inspiration came from Tableau dashboards and examples found in data visualisation literature. To guarantee readability and prevent deceptive interpretations, the choice of colour was carefully researched. Aggregated data may obscure individual-level differences, which is one drawback of some visualisations. Furthermore, because the dataset is generated, it could not accurately represent patterns found in the real world. Nonetheless, the visualisations were created to facilitate interpretation and efficiently convey important findings.